University of Cape Town

Time Series

Portfolio

Insert some title here

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Overview

Overview

Welcome to my portfolio for the Department of Statistical Sciences. This document showcases my learning progression, statistical understanding, computational applications, and reflections across the course.

Portfolio Structure

- Reflective Commentary (1000–2000 word synthesis)
- AI Use Log (Record of generative AI assistance)
- Course Section 1 (Tutorials, model fitting, and reflections)
- Course Section 2 (Tutorials, model fitting, and reflections)
- Substantial Projects (Four major analytical projects)
- Learning Journal (Chronological progress tracking)
- Appendix (Reproducible code and dataset links)[cite: 1]

Reflective Commentary

Learning Achievements

[cite: 1]

Methodological Challenges

[cite: 1]

Ethical and Reproducibility Considerations

[cite: 1]

Future Areas for Development

[cite: 1]

AI Use Log

Date	Tool Used	Prompt / Task Description	Link to Conversation	Verification & Changes Made
YYYY-MM-DD	ChatGPT / Claude	Debugging R code for model fit	Link	Checked output against manual calculations[cite: 1]

Course Section 1

Data Statement

[cite: 1]

Teaching Session Summaries

[cite: 1]

Worked Tutorials & Software Implementations

[cite: 1]

Topic Exploration

[cite: 1]

Course Section 2

Data Statement

[cite: 1]

Teaching Session Summaries

[cite: 1]

Worked Tutorials & Software Implementations

[cite: 1]

Topic Exploration

[cite: 1]

Substantial Projects

Overview

This section contains four substantial projects spanning Section 1 and Section 2[cite: 1].

Project	Section	Project Type	Focus Area	Link
Project 1	Section 1	Simulation Study	Model Comparison	Summary[cite: 1]
Project 2	Section 1	R Package Implementation	Computational Programming	Summary[cite: 1]
Project 3	Section 2	Shiny Application	Interactive Visualization	Summary[cite: 1]
Project 4	Section 2	Exploratory Data Analysis	Real-World Data	Summary[cite: 1]

Project 1

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Project 2

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]

- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Project 3

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Project 4

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Learning Journal

This journal documents my continuous learning progress throughout the course[cite: 1].

- View Chronological Entries[cite: 1]

Entry 1: YYYY-MM-DD

- **Session / Topic Covered:**[cite: 1]
- **What I Did:**[cite: 1]
- **What I Learned:**[cite: 1]
- **Difficulties / Open Questions:**[cite: 1]
- **What to Do Next:**[cite: 1]
- **Links:** Link to Section Work | Link to AI Log[cite: 1]

Appendix

R Code & Scripts

- `scripts/simulation-study.R`
- `scripts/shiny-app/app.R`

Data Access & Repositories

- Raw Data Files & Data Access Instructions[cite: 1]

Computational Notebooks

- Extended Analysis Reports[cite: 1]